Implicit Functions (implicit differentiation)
&
Implicit Function Theorem

F18CE Week 3

1. Motivation

If we define the function $f(x,y) = x^2 + y^2 - 1$, then the equation $f(x,y) = 0$ is the unit circle. However, there is no way we can represent the unit circle as the graph of a function of one variable $y = g(x)$ because for each choice of $x \in (-1, 1)$, there are two choices of y, namely $\pm\sqrt{1 - x^2}$.

Implicit Function

f(x,y) = 0

Explicit Function

y = f(x)

However, it is possible to represent *part* of the circle as the graph of a function of one variable. If we let $g_1(x) = \sqrt{1 - x^2}$ for $-1 \leq x \leq 1$, then the graph of $y = g_1(x)$ provides the upper half of the circle. Similarly, if $y = g_2(x) = -\sqrt{1 - x^2}$, then the graph gives the lower half of the circle.

The purpose of the implicit function theorem is to tell us the existence of functions like $g_1(x)$ and $g_2(x)$, even in situations where we cannot write down explicit formulas.

2. Example 3.4 & Example 3.5

Example. The equation $y - x - \frac{1}{2}\sin y = 0$ can be easily written for $x(y) = y - \frac{1}{2}\sin y$ but cannot be written explicitly for $y(x)$.

Example. The equation $\ln\left(x^2 + y^2\right) = \tan^{-1}\left(\frac{y}{x}\right)$ cannot be written explicitly for either variable.

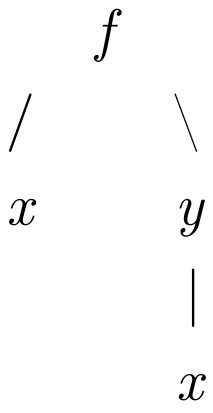
Important Remark

- Implicit function theorem is not implicit differentiation formula.
- One is used to determine the existence of solution and one is for finding the implicit differentiation.



3. Implicit differentiation formula

3. Suppose $y(x)$ is a solution function for $f(x,y)=0$.



Now we differentiate $f(x,y)=0$ with respect to x.

$$f(x,y)=0 \Rightarrow \frac{df}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\times\frac{dy}{dx}=0 \Rightarrow \boxed{\frac{dy}{dx}=-\frac{f_x}{f_y}}$$

This formula is called the implicit differentiation formula.

4. Example 3.3 & Example 3.6

Let $y = \sqrt{1-x^2}$ (from example $f(x,y) = x^2 + y^2 - 1$). Then

y is a solution for $f(x,y) = 0$.

We can find y' by using (a) explicit differentiation; or

by using (b) the implicit differentiation formula.

(a) $y = \sqrt{1-x^2} \Rightarrow y' = \dfrac{-x}{\sqrt{1-x^2}}$ (b) $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y} = -\dfrac{2x}{2y} = -\dfrac{x}{y} = -\dfrac{x}{\sqrt{1-x^2}}$

5. Example 3.9

Suppose $y(u,v) : \mathbb{R}^2 \to \mathbb{R}$ is a function defined implicitly by

$$y^2 + v^7 + \frac{5vy}{u^2+1} = u\,.$$

Find $\frac{\partial y}{\partial u}$ and $\frac{\partial y}{\partial v}$.

Solution. Write $(u^2+1)y^2 + (u^2+1)v^7 + 5vy = u(u^2+1)\,.$

Partially (and implicitly) differentiate the above with respect to u:

$$\begin{array}{ccc} y & & 2y\frac{\partial y}{\partial u}(u^2+1) + 2uy^2 + 2uv^7 + 5v\frac{\partial y}{\partial u} = 3u^2+1 \\ /\ \backslash & & \Rightarrow \frac{\partial y}{\partial u} = \frac{3u^2+1-2uy^2-2uv^7}{2u^2y+2y+5v} \\ u \quad v & & \end{array}$$

Partially (and implicitly) differentiate the above with respect to v:

$$\left(2y\frac{\partial y}{\partial v}+7v^6\right)(u^2+1)+5y+5v\frac{\partial y}{\partial v}=0$$

$$\Rightarrow \frac{\partial y}{\partial v}=-\frac{7v^6(u^2+1)+5y}{2u^2y+2y+5v}$$

6. Definition 3.1. C^1-class function

Let $f : D_f \to \mathbb{R}$, where $D_f \subseteq \mathbb{R}^n$ be a function. Then f is called a C^1-class function if all of its (first order) partial derivatives exist and are continuous on its domain.

7. Definition 3.2. Solution function

(a) A solution of an equation $f(x,y)=0$ is any pair of numbers, say (a,b), for which $f(a,b)=0$.

(b) We say that a function $y(x)$ is a solution function if $f(x,y(x))=0$ for all $x \in D_y$, where D_y denotes the domain of $y(x)$.

(c) Similarly, $x(y)$ is a solution if $f(x(y),y)=0$ for all $y \in D_x$, where D_x denotes the domain of $x(y)$.

8. Example of solution function

Revisit our first example $f(x,y) = x^2 + y^2 - 1 = 0$. Then

(a) $(1,0)$ is a solution because $f(1,0) = 1^2 + 0^2 - 1 = 0$.

(b) $(0,-1)$ is also a solution (we have infinitely many solutions).

(c) $y(x) = \sqrt{1-x^2}$ is a solution function because $f(x,y) = x^2 + (1-x^2) - 1 = 0$.

(d) $y(x) = -\sqrt{1-x^2}$ is also a solution function.

(e) $x(y) = \sqrt{1-y^2}$ is also a solution function.

Remark: The solution function may not exist, and if it exists, it may not be unique.

So we normally we consider a solution function at a specific location, if it exists.

Question

Given an implicit function f(x,y) = 0. How do we know if there exist a solution function (near a point)?

9. Theorem 3.7 (functions of 2-variable)

Let $f : D_f \to \mathbb{R}, D_f \subseteq \mathbb{R}^2$ be a C^1-class function.

If there exists a point $(a,b) \in D_f$ such that $f_y(a,b) \neq 0$.

Then there exists a solution function $y(x)$ near (a,b) with $y(a) = b$.

Remark

Moreover, there exists an $\varepsilon > 0$ and a unique solution function $\tilde{y}$ with $\left|\tilde{y} - b\right| < \varepsilon$ such that $\tilde{y}(a) = b$.

Remark: Referring to slide #6, since $f(x, y(x)) = 0$,

it follows that $\frac{d}{dx} y(x) = -\frac{f_x(x,y(x))}{f_y(x,y(x))}$.

10. How to apply the theorem?

Let $f : D_f \to \mathbb{R}, D_f \subseteq \mathbb{R}^2$.

Step 1: Check that f is a C^1-class function.

Step 2: Find a solution point, i.e., $(a,b) \in D_f$ such that $f(a,b) = 0$.

Step 3: Verify that $f_y(a,b) \neq 0$.

Conclusion: There exists a solution function $y(x)$ near (a,b) with $y(a) = b$.

Note that

- If one of the three conditions are failed to comply, it does not mean the solution function does not exist. We do not know whether the solution function exists or not!

11. Revisit Example 3.6

Revisit the example of $f(x,y) = x^2 + y^2 - 1$. Then we see that

1. $\frac{\partial f}{\partial x} = 2x$ and $\frac{\partial f}{\partial y} = 2y$ exist and are continuous (everywhere).
2. $f(0,1) = 0$ (we could choose $f(0,-1) = 0$ but then $y(x)$ would be different)
3. $f_y(0,1) = 2 \neq 0$.

We conclude that there exists a solution function $y(x)$ near $(0, 1)$ such that $y(0) = 1$.

12. Theorem 3.8. Implicit function theorem

Let $f(x,y):\mathbb{R}^{m+n}\to\mathbb{R}^n$ be a function such that

$$f(x,y)=\begin{pmatrix} f_1(x,y)\\ \vdots \\ f_n(x,y)\end{pmatrix},$$

where $x=(x_1,...,x_m)\in\mathbb{R}^m$ (independent) and

$y=(y_1,...,y_n)\in\mathbb{R}^n$ (dependent).

Theorem 3.8 (Generalisation)

Let $f = \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} : \mathbb{R}^{m+n} \to \mathbb{R}^n$ be a function, where $f_i : \mathbb{R}^{m+n} \to \mathbb{R}$

are C^1-class functions for all $i = 1, \ldots, n$.

If there exists a point $(\mathbf{a}, \mathbf{b}) \in \mathbb{R}^{m+n}$ $(\mathbf{a} \in \mathbb{R}^m, \mathbf{b} \in \mathbb{R}^n)$ such that

$D_{\mathbf{y}} f(\mathbf{a}, \mathbf{b})$ is invertible. Then there exists a solution function

$\mathbf{y}(\mathbf{x}) : \mathbb{R}^m \to \mathbb{R}$ defined for $\mathbf{x} : \mathbb{R}^n \to \mathbb{R}$ near $(\mathbf{a}, \mathbf{b})$ with $\mathbf{y}(\mathbf{a}) = \mathbf{b}$.

Remark

$D_{\mathbf{y}} f(\mathbf{a}, \mathbf{b})$ is the matrix Y in the total derivatives:

$$Df(\mathbf{a}, \mathbf{b}) = \left(\begin{array}{ccc|ccc} \frac{\partial f_1}{\partial x_1}(\mathbf{a}, \mathbf{b}) & \dots & \frac{\partial f_1}{\partial x_m}(\mathbf{a}, \mathbf{b}) & \frac{\partial f_1}{\partial y_1}(\mathbf{a}, \mathbf{b}) & \dots & \frac{\partial f_1}{\partial y_m}(\mathbf{a}, \mathbf{b}) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1}(\mathbf{a}, \mathbf{b}) & \dots & \frac{\partial f_n}{\partial x_m}(\mathbf{a}, \mathbf{b}) & \frac{\partial f_n}{\partial y_1}(\mathbf{a}, \mathbf{b}) & \dots & \frac{\partial f_n}{\partial y_m}(\mathbf{a}, \mathbf{b}) \end{array} \right) = (X \mid Y)$$

What exam question will normally test you?

Let $f(x,y,z):\mathbb{R}^3 \to \mathbb{R}$ be a function. Suppose that $f_x, f_y, f_z \in C^1$ and $\exists (a,b,c) \in \mathbb{R}^3$ э $f(a,b,c)=0$. Then

(a) $f_x(a,b,c) \neq 0 \Rightarrow \exists$ solution function $x = x(y,z)$ near to (a,b,c) s.t. $x(b,c) = a$.

(b) $f_y(a,b,c) \neq 0 \Rightarrow \exists$ solution function $y = y(x,z)$ near to (a,b,c) s.t. $y(a,c) = b$.

(c) $f_z(a,b,c) \neq 0 \Rightarrow \exists$ solution function $x = z(x,y)$ near to (a,b,c) s.t. $z(a,b) = c$.

13. Example 3.10

Consider the equation $z^2 + 2xy + xz = 0$.

(a) Does the implicit function theorem apply, and

can we solve for $z = z(x, y)$ near the points

(i) $(x, y, z) = (1, -1, 1)$? (ii) $(x, y, z) = (-2, -\frac{1}{4}, 1)$?

(b) For (a)(ii), can we obtain a solution function for $y = y(x, z)$?

Solution. (a) Let $f(x,y,z) = z^2 + 2xy + xz$.

Now, $f_x = 2y + z, f_y = 2x, f_z = 2z + x$.

For (i)	For (ii)
(1) f is a C^1-class function. (2) $f(1,-1,1) = 1-2+1 = 0$. (3) $f_z(1,-1,1) = 2+1 = 3 \neq 0$.	(1) f is a C^1-class function. (2) $f(-2,-\frac{1}{4},1) = 1+1-2 = 0$. (3) but $f_z(-2,-\frac{1}{4},1) = 0$.
Conclusion: Solution function exists.	No conclusion, theorem not applicable.

(b)

For (ii)
(1) f is a C^1-class function. (2) $f(-2,-\frac{1}{4},1) = 1+1-2 = 0$. (3) $f_y(-2,-\frac{1}{4},1) = -4 \neq 0$.
Conclusion: There exists a solution function $y = y(x, z)$ near that point $(-2, -\frac{1}{4}, 1)$.

14. Extra Example (a)

Let $u(w), v(w)$ be functions in w. Consider

$$u^2 + uv - w - vw = 0$$
$$u + u^2 v + v^2 w - 3 = 0.$$

Use the implicit function theorem to determine whether there exists a C^1-class solution function $(u(w), v(w))$ in the neighbourhood of the solution point $w = 1$.

Solution. Let $f(w,u,v) = \begin{pmatrix} u^2 + uv - w - vw \\ u + u^2v + v^2w - 3 \end{pmatrix}$, where w is an independent variable and u, v are dependent. Let $y = (u,v) : \mathbb{R} \to \mathbb{R}^2$ and $u(w)$, $v(w)$ are functions in w.

(1) For $f_1 = u^2 + uv - w - vw = 0$, $f_2 = u + u^2v + v^2w - 3 = 0$, we see that both of these functions are continuous everywhere, their first order partial derivatives exist

$$\frac{\partial f_1}{\partial u} = 2u + v, \qquad \frac{\partial f_1}{\partial v} = u - w, \qquad \frac{\partial f_1}{\partial w} = -1 - v$$
$$\frac{\partial f_2}{\partial u} = 1 + 2uv, \qquad \frac{\partial f_2}{\partial v} = u^2 + 2vw, \qquad \frac{\partial f_2}{\partial w} = v^2$$

and they are all continuous.

(2) We see that $f(1,1,1) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

(3) $\det \begin{pmatrix} 2u+v & u-w \\ 1+2uv & u^2+2vw \end{pmatrix}\Big|_{u=v=w=1} = \det \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix} = 9 \neq 0.$

Therefore, there exists a C^1-class solution

$(u(w), v(w))$ in the neighbourhood of the solution point $w = 1$.

15. Extra Example (b)

Given

$$\begin{aligned} f(x,y,z) &= x\sin y + yz + z^3 + 2z &&= 0 \\ g(x,y,z) &= x + e^{xy} - yz^2 &&= 0 \end{aligned}$$

What does the implicit function theorem say about the existence of C^1-class solution function $(x(z),\ y(z))$ near $z = 0$?

Let h be defined as $h(x,y,z) = \begin{pmatrix} f(x,y,z) \\ g(x,y,z) \end{pmatrix}$, where z is independent, and x and y are dependent.

(1) Note that f and g are continuous, and their first partial derivatives exist

$$\frac{\partial f}{\partial x} = \sin y, \qquad \frac{\partial f}{\partial y} = x\cos y + z, \qquad \frac{\partial f}{\partial z} = y + 3z^2 + 2$$
$$\frac{\partial g}{\partial x} = 1 + ye^{xy}, \qquad \frac{\partial g}{\partial y} = xe^{xy} - z^2, \qquad \frac{\partial g}{\partial z} = -2yz$$

and they are all continuous.



(2) We see that $f(-1,0,0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.



(3) $$\det \begin{pmatrix} \sin y & x\cos y + z \\ 1 & xe^{xy} - z^2 \end{pmatrix}\Bigg|_{(x,y,z)=(-1,0,0)} = \det \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} = 1 \neq 0.$$

Thus, by virtue of the implicit function theorem, there exists a C^1-class solution function $(x(z),\ y(z))$ near $z = 0$.

16. Extra example (c): Final exam question 2023/24

Consider the equation $2x \sin y + x(e^z - 1) = 0$. Use the Implicit Function Theorem to determine whether it is possible to define an implicit function $z(x, y)$ in the neighbourhood of $(x, y) = (2, 0)$ with $z(2, 0) = 0$. Justify your answer. (6 marks)

Solution. Let $f(x,y,z) = 2x\sin y + x(e^z - 1)$.

1. Note that f and all first partial derivatives exist and continuous everywhere

$$f_x = 2\sin y + e^z - 1,\ \ f_y = 2x\cos y,\ \ f_z = xe^z\,.$$

2. We see that $f(2,0,0) = 0$ so that $(2,0,0)$ is a solution point.

3. $f_z(2,0,0) = 2e^0 = 2 \neq 0$.

So the validity of the implicit function theorem holds, guaranteeing the existence of the required function $z(x,y)$.

Verify that

$$2x\sin y + x(e^z - 1) = 0 \Rightarrow e^z = x - 2x\sin y \Rightarrow z = \ln\Big(x - 2x\sin y\Big)$$

for $x > 2x\sin y$ where $(2,0,0)$ fulfils this condition.

17. Extra example (d)

Let $f(u,v)$ be a differentiable function of two variables, and $z = z(x,y)$ be a differentiable function defined implicitly by

$$f(xz, yz) = f(xz(x,y), yz(x,y)) = 0\,.$$

Given that $xf_u + yf_v \neq 0$, show that $xz_x + yz_y = -z$.

Solution. Partially differentiate $f(u,v)=f(xz,yz)=0$ with respect to x and y.

Note that $u_x=z+xz_x, u_y=xz_y, v_x=yz_x, v_y=z+yz_y$. If $xf_u+yf_v\neq 0$, then

$$0=f_u u_x+f_v v_x=\left(z+xz_x\right)f_u+yz_x f_v \Rightarrow z_x=-\frac{zf_u}{xf_u+yf_v}$$

$$0=f_u u_y+f_v v_y=xz_y f_u+\left(z+yz_y\right)f_v \Rightarrow z_y=-\frac{zf_v}{xf_u+yf_v}$$

$$\therefore xz_x+yz_y=-\frac{xzf_u}{xf_u+yf_v}-\frac{yzf_v}{xf_u+yf_v}=-z\frac{xf_u+yf_v}{xf_u+yf_v}=-z.$$

If $z=0$, then obviously $xz_x+yz_y=-z$ holds true.

Now if $xf_u + yf_v = 0$ and $z \neq 0$, then

$$\begin{aligned} 0 &= zf_u + \Big(xf_u + yf_v\Big)z_x \\ 0 &= zf_v + \Big(xf_u + yf_v\Big)z_y \end{aligned} \Rightarrow zf_u = 0 = zf_v \Rightarrow f_u = 0 = f_v\,.$$

This means that the function $f(u,v) = C =$ constant.

From equation $f\Big(xz(x,y), yz(x,y)\Big) = 0$, let $F(x,y,z) = f(xz,yz)$. Then $F_z = 0$.

This does not guarantee the existence of an implicit function z in terms of x and y.